Retail Sales & Profitability Analysis

STUDENT PROJECT BRIEF • Python Data Analysis • Suggested duration: 2 hours

1. Business Scenario

A retail business operates three branches and sells products across several categories. Management wants to understand sales, profit, discount effects, and differences between branches and customer types.

2. Project Mission

Create a retail performance analysis that distinguishes high sales from healthy profitability and recommends actions to improve results.

3. Learning Objectives

- Inspect a real-world-style dataset and explain its structure and limitations.
- Use NumPy and Pandas for data types, missing values, duplicates, derived columns, grouping, and aggregation.
- Use Matplotlib, Seaborn, and/or Plotly to communicate patterns with appropriate charts.
- Translate findings into practical, evidence-based business recommendations.
- Build a simple interactive Streamlit dashboard for a non-technical stakeholder.

4. Provided Data

retail_sales_raw.csv — Transaction-level sales data with date, branch, category, product, quantity, price, discount, sales, cost, profit, payment method, and customer type.

5. Required Tasks

- **Task A — Understand the data:** load the file(s), inspect shape, columns, data types, sample rows, and summary statistics.
- **Task B — Clean the data:** Check missing values, duplicated Transaction_ID values, invalid dates, non-positive quantities/prices, discount values outside the expected range, and inconsistencies between Sales_EGP, Cost_EGP, and Profit_EGP. Document any recalculations.
- **Task C — Explore and analyze:** answer the business questions below using suitable groupby/aggregation and at least one time-based analysis where dates are available.
- **Task D — Visualize:** create at least 4 meaningful charts. Add clear titles, labels, units, and readable formatting.
- **Task E — Recommend:** write 3–5 prioritized recommendations. For each, cite the finding that supports it and mention any limitation.
- **Task F — Build the dashboard:** use Streamlit to present KPIs, charts, and at least two interactive filters where appropriate.

6. Business Questions to Answer

- How do sales, profit, and profit margin change over time?
- Which branches, categories, and products contribute most to sales and profit?
- Are high discounts associated with lower profit margins?
- Which payment methods and customer types have the highest sales or profit?
- What actions could improve profitability without unnecessarily reducing sales?

7. KPI Definitions / Analysis Focus

Total sales; total cost; total profit; profit margin; transaction count; average transaction value; average discount.

8. Suggested Visualizations

Monthly sales and profit trend; branch comparison; category/product contribution; discount vs profit margin; customer type and payment-method comparisons.

9. Important Project-Specific Rule

A product or branch with the highest revenue is not necessarily the most profitable. Show both sales and profit measures.

10. Recommended Workflow (120 minutes)

Time	Activity	Expected output
0–10 min	Understand scenario and inspect dataset(s)	Data dictionary / initial observations
10–30 min	Clean data and document decisions	Cleaned dataframe(s)
30–60 min	Calculate KPIs and explore patterns	Analysis tables and findings
60–85 min	Build and refine visualizations	At least 4 charts
85–110 min	Build Streamlit dashboard	Working dashboard with filters
110–120 min	Summarize recommendations and prepare demo	Short presentation

11. Deliverables

- A Jupyter Notebook (.ipynb) or Python analysis script with readable, commented code.
- A working Streamlit app (app.py) with KPIs, visualizations, and filters.
- A cleaned dataset or a reproducible cleaning pipeline; do not overwrite the original raw data.
- A one-page executive summary containing 3–5 findings, 3 recommendations, and limitations.
- A short 3–5 minute presentation/demo explaining the business problem, analysis, dashboard, and recommendations.

12. Submission and Technical Requirements

- Use Python, Pandas, NumPy, and at least one of Matplotlib, Seaborn, or Plotly.
- Use Streamlit for the dashboard. Include a requirements.txt file and clear run instructions.
- Make the project reproducible: use relative paths, descriptive variable names, and avoid hard-coded machine-specific paths.
- Never claim correlation proves causation. Explain assumptions and limitations.
- Submit all code and supporting files in one clearly named folder or ZIP.

13. Assessment Rubric

Criterion	Weight
Data understanding and cleaning	20%
Analysis and business metrics	25%
Visual storytelling and chart choice	20%
Streamlit dashboard usability	20%
Recommendations and presentation	15%
Total	100%

14. Viva / Discussion Questions

- What was the most important data-quality issue you found, and how did you handle it?
- Which KPI best answers the business problem, and why?
- Why did you choose each chart, and what could make it misleading?
- Which finding is strongest, and what additional data would validate it?
- If you had one more day, what would you improve in the analysis or dashboard?

15. Final Submission Checklist

- I checked the shape, data types, missing values, and duplicates.
- I documented cleaning decisions and kept raw data unchanged.
- I validated KPI formulas and checked for division-by-zero or double-counting issues.
- My charts have clear titles, labels, units, and readable scales.
- My dashboard runs from the provided instructions.
- My recommendations are linked to findings and acknowledge limitations.